Chapter 22

Logarithms

counting the multiplications



$= 8$

$$\log_2 8 = 3$$

Powers ask: **multiply 2 by itself 3 times, what do you get?** (8). Logarithms ask the question the other way round: **how many 2s must I multiply to get 8?** (3). That's all a log is: **a count of multiplications**, or equivalently, **the missing power**.

Logs are the undo button for exponentials (Chapter 21), so they're how you answer "how long until...?" questions. Each log rule is just an index rule (Chapter 9) wearing a disguise, and this chapter shows them side by side.

Goes with **Lecture 7** (logarithms, laws of logs, natural log, Questions 1-3).

The question a log asks

RULE: definition of a logarithm

$y = \log_b x \iff b^y = x$. A log **is a power**: the power you raise b to, to get x.

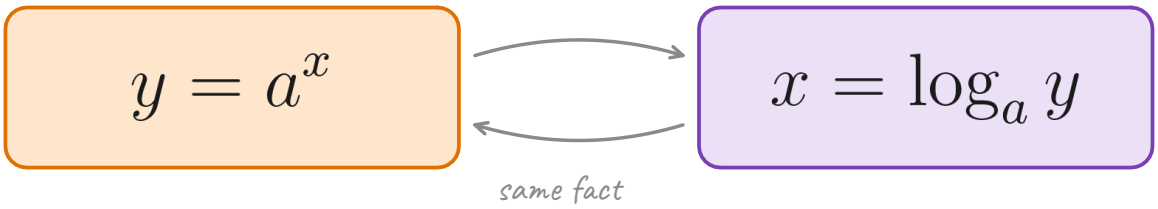
The base b is positive and not 1. The number x must be **positive**: there is no log of 0 or of a negative number.

$\log_b b = 1$ (since $b^1 = b$) and $\log_b 1 = 0$ (since $b^0 = 1$).

Read $\log_b N$ as "**log to the base b of N**", and hear it as the question: **what power of b gives N?** The small number b is the **base**: the thing being multiplied.

$\log_2 8$	*how many 2s make 8?*	$2 \times 2 \times 2 = 8$	**3**
$\log_{10} 100$	*how many 10s make 100?*	$10 \times 10 = 100$	**2**
$\log_4 16$	*how many 4s make 16?*	$4 \times 4 = 16$	**2**
$\log_{10} 1000000$	*how many 10s make a million?*	10^6	**6**
$\log_5 5$	*how many 5s make 5?*	5	**1**
$\log_7 1$	*how many 7s make 1?*	$7^0 = 1$	**0**

The same fact in two languages. Every log statement is a power statement in disguise. Lecture 7's translation rule:



A handy chant: "the base stays the base, and the log is the power".

Logs are the undo button for powers

RULE: logs and powers undo each other

$\log_b(b^x) = x$ and $b^{\log_b x} = x$ (for $x > 0$).

Base e: $\ln x$ means $\log_e x$. So $\ln(e^x) = x$, $e^{\ln x} = x$, $\ln e = 1$, $\ln 1 = 0$.

Base 10: $\log x$ (the calculator's log button) means $\log_{10} x$. So $\log 10 = 1$, $\log 1 = 0$.

In Chapter 14 every operation had an undo button: + undoes −, × undoes ÷, squaring undoes square-rooting. **"Raise 10 to the power" and "take $\log_{10}$" undo each other**:

$$\log_{10}(10^x) = x$$ ← *power first, then log: back to x*

$$10^{\log_{10} x} = x$$ ← *log first, then power: back to x*

Your calculator has two log buttons. **log** means base 10 (the **common log**); **ln** means base e (the **natural log**, Lecture 7: we write ln rather than $\log_e$).

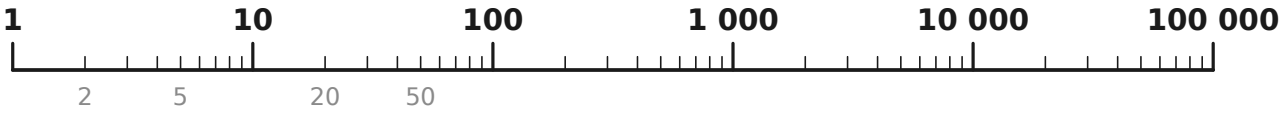
Logs of in-between numbers are decimals. 50 is between 10 and 100, so $\log_{10} 50$ is between 1 and 2:



What you can't log: $\log 1 = 0$ (since $10^0 = 1$), numbers between 0 and 1 have **negative** logs ($\log 0.01 = -2$), and 0 or negative numbers have **no log at all**: no power of 10 is ever 0 or negative (the curve of 10^x never reaches the x-axis). So the domain of log is positive numbers only.

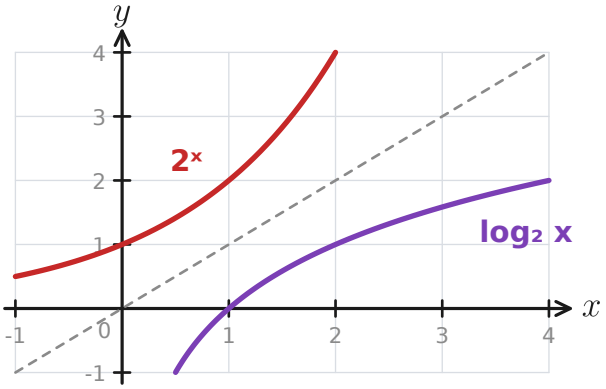
Log scales: a ruler where every step is ×10

On an ordinary ruler every step **adds** the same amount. On a **log scale** every step **multiplies** by the same amount. That squashes enormous ranges onto one page:



equal gaps from 1 to 10, 10 to 100, 100 to 1000… and 2 sits 30% of the way to 10, because log 2 ≈ 0.301

You meet log scales in **decibels** (sound), **pH** (acidity) and the **Richter scale** (earthquakes): one step up means 10 times more intense. They also straighten out exponential curves: plot 10^x on a log axis and you get a straight line.



The graph of log is the graph of the power, reflected in the dashed line y = x (because they undo each other: inputs and outputs swap).

It rises ever more slowly and never exists for x ≤ 0.

The log laws are the index laws in disguise

RULE: the laws of logarithms (same base throughout)

$\log_b(xy) = \log_b x + \log_b y$ $\log_b \dfrac{x}{y} = \log_b x - \log_b y$ $\log_b x^n = n \log_b x$

Special cases: $\log_b \dfrac{1}{x} = -\log_b x$ and $\log_b \sqrt{x} = \frac{1}{2}\log_b x$.

The laws only combine logs with the **same base**: $\log_2 x + \log_3 y$ can't be combined (change base first).

There is **no law** for $\log(x+y)$ or $\log x \times \log y$.

A log **is** a power, so the three index laws of Chapter 9 turn straight into three log laws. Read across each row: the index law on the left, the log law on the right, and a number test.

$$10^m \times 10^n = 10^{m+n} \quad \rightarrow \quad \log(ab) = \log a + \log b$$

$\log(100 \times 1000) = \log 10^5 = 5 = 2 + 3$

multiply inside → add outside

$$\frac{10^m}{10^n} = 10^{m-n} \quad \rightarrow \quad \log \frac{a}{b} = \log a - \log b$$

$\log \dfrac{1000}{10} = \log 100 = 2 = 3 - 1$

divide inside → subtract outside

$$(10^m)^n = 10^{mn} \quad \rightarrow \quad \log a^n = n \log a$$

$\log 1000^2 = \log 10^6 = 6 = 2 \times 3$

power inside → multiply in front

Traps (test with numbers!): $\log(a+b)$ is **not** $\log a + \log b$ (try $\log(10+10) = 1.30$, but $1+1=2$). And $\dfrac{\log a}{\log b}$ is **not** $\log \frac{a}{b}$.

(Lecture 7's example for $\log a^n = n \log a$ is a copy of the first law's example. A real one: $\log_{10} a^3 = 3 \log_{10} a$.)

PROCEDURE: combining several logs into one

1. Check every log has the same base.
2. Move every number in front up into a power: $n \log x = \log x^n$.
3. Every $+$ log goes on top of one fraction, every $-$ log goes underneath: $\log A + \log B - \log C = \log \dfrac{AB}{C}$.
4. Simplify the fraction inside the log.

$$2\log 3 + \log 4 - \log 6 = \log 9 + \log 4 - \log 6$$

step 2: 2 log 3 = log 3^2

$$= \log \frac{9 \times 4}{6} = \log 6$$ ← steps 3 and 4

$$\ln x - 3\ln y + \tfrac{1}{2}\ln z = \ln \frac{x\sqrt{z}}{y^3}$$ ← powers first; minus terms underneath

$$\log \frac{1}{100} = \log 1 - \log 100 = 0 - 2 = -2$$ ← log(1/a) = −log a

Lecture 7, Question 1: pulling a log apart

Express $\log_{10} \frac{a^2 b^3}{100\sqrt{c}}$ as separate logs of a, b and c. One law per line; do the division first, because it's the "outermost" operation. First rewrite the root as a power (Chapter 9): $\sqrt{c} = c^{1/2}$.

$$\log \frac{a^2 b^3}{100 c^{1/2}} = \log(a^2 b^3) - \log(100 c^{1/2})$$ ← divide → subtract

$$= \log a^2 + \log b^3 - \left(\log 100 + \log c^{1/2}\right)$$ ← multiply → add (keep the bracket!)

$$= 2\log a + 3\log b - (2 + \tfrac{1}{2}\log c)$$ ← powers come to the front; log 100 = 2

$$= 2\log a + 3\log b - 2 - \tfrac{1}{2}\log c$$ ← the minus reaches both terms

Test with numbers: $a = 3$, $b = 5$, $c = 7$: both sides give 0.6286 ✓.

How many doublings? Solving with logs

PROCEDURE: solving an equation with the unknown in the power

1. Get the power on its own on one side: $b^{(\dots)} = N$.
2. Take logs of **both** sides (log or ln; use ln when the base is e).
3. Use the power law to bring the unknown down: $(\dots)\log b = \log N$.
4. Solve the ordinary equation that's left. If the unknown is in two powers, expand, collect the x terms and factorise out x.

When the unknown is **in the power**, no amount of adding or dividing will get it down. Lecture 7: **take logs of both sides** (allowed: do the same to both sides), then use the power law to bring it to the front:

$$2^x = 8$$

$$\log 2^x = \log 8 \quad \longleftarrow \text{log both sides}$$

$$x \log 2 = \log 8 \quad \longleftarrow \text{the power law lifts } x \text{ down}$$

$$x = \frac{\log 8}{\log 2} = \frac{0.903}{0.301} = 3 \quad \longleftarrow \div \log 2 \text{ (the slide's 0.90309)}$$

Now one you couldn't guess: how many years until £1000 at 5% doubles to £2000?

$$1000 \times 1.05^t = 2000 \quad \longleftarrow \text{Chapter 21's formula}$$

$$1.05^t = 2 \quad \longleftarrow \div 1000$$

$$t = \frac{\log 2}{\log 1.05} \approx 14.2 \quad \longleftarrow \text{log both sides; power law; } \div \log 1.05$$

So 15 whole years (after 14 years it's only £1979.93). Change of base (Lecture 7) is the same trick: $\log_5 15 = \frac{\log 15}{\log 5} = \frac{1.1761}{0.6990} = 1.6825$, which lets a base-10 calculator do any base.

RULE: change of base

$\log_a N = \dfrac{\log N}{\log a} = \dfrac{\ln N}{\ln a}$: any base can be worked out with the log or ln button.

Example: solve $3^{2x-1} = 7$ (3 d.p.).

$$(2x-1)\log 3 = \log 7 \quad \leftarrow \text{steps 2 and 3: log both sides, power down}$$

$$2x - 1 = \frac{\log 7}{\log 3} = 1.7712 \quad \leftarrow \div \log 3$$

$$x = \frac{2.7712}{2} = 1.386 \quad \leftarrow +1, \text{ then } \div 2$$

Example: the unknown on both sides: $2^{x+1} = 3^x$ (3 d.p.).

$$(x+1)\log 2 = x \log 3 \quad \leftarrow \text{log both sides}$$

$$x\log 2 + \log 2 = x \log 3 \quad \leftarrow \text{expand}$$

$$\log 2 = x(\log 3 - \log 2) \quad \leftarrow x \text{ terms together; factorise out } x$$

$$x = \frac{\log 2}{\log 3 - \log 2} = \frac{0.3010}{0.1761} = 1.710$$

Example: find k if $50 = 100e^{-5k}$ (3 d.p.).

$$e^{-5k} = 0.5 \quad \leftarrow \div 100 \text{ to get the power alone}$$

$$-5k = \ln 0.5 \;\Rightarrow\; k = \frac{\ln 0.5}{-5} = 0.139 \quad \leftarrow \ln \text{ both sides (base } e)$$

Example: a substance has a half-life of 5 years (Chapter 21). How long until 80 g has decayed to 30 g?

$$80\left(\tfrac{1}{2}\right)^{t/5} = 30 \;\Rightarrow\; 0.5^{t/5} = 0.375 \quad \leftarrow \div 80$$

$$\frac{t}{5} = \frac{\log 0.375}{\log 0.5} = 1.415 \;\Rightarrow\; t \approx 7.08 \text{ years}$$

Example: Chapter 17's question: the smallest n with $2^n > 1001$.

$$n \log 2 > \log 1001 \;\Rightarrow\; n > \frac{\log 1001}{\log 2} = 9.97$$

log 2 > 0, so the > stays

So $n = 10$, the next whole number, as found by trial in Chapter 17.

Solving equations that contain logs

PROCEDURE: solving log equations

1. Use the laws to collect the logs into **one** log on one side (same base): $\log_b(\dots) = c$.
2. Rewrite as a power, by the definition: $\log_b(\dots) = c \;\Rightarrow\; (\dots) = b^c$. (For $\ln$, $b = e$; for $\log$, $b = 10$.)
3. Solve the equation that's left.
4. **Check every answer in the original**: the inside of each log must be positive. Reject any root that makes a log of 0 or of a negative number.

$$\log_2 x = 5 \;\Rightarrow\; x = 2^5 = 32$$ ← *definition: the log is the power*

$$\ln x = 2 \;\Rightarrow\; x = e^2 \approx 7.389$$ ← *base e*

$$\log_3(2x+1) = 2 \;\Rightarrow\; 2x+1 = 9 \;\Rightarrow\; x = 4$$

$3^2 = 9$

Example: solve $\log x + \log(x+3) = 1$ (base 10).

$$\log\big(x(x+3)\big) = 1$$ ← *step 1: add logs = log of the product*

$$x(x+3) = 10^1 = 10$$ ← *step 2: rewrite as a power*

$$x^2 + 3x - 10 = 0 \;\Rightarrow\; (x+5)(x-2) = 0$$

step 3: a quadratic

$$x = 2 \quad (x = -5 \text{ rejected})$$ ← *step 4: log(−5) doesn't exist*

Check: $\log 2 + \log 5 = \log 10 = 1$ ✓. The root $x = -5$ solves the quadratic but not the original, because $\log(-5)$ has no value.

Example: solve $\log_2 x - \log_2(x-2) = 3$.

$$\log_2 \frac{x}{x-2} = 3 \;\Rightarrow\; \frac{x}{x-2} = 2^3 = 8$$ ← subtract logs = log of the quotient

$$x = 8x - 16 \;\Rightarrow\; x = \frac{16}{7}$$ ← × (x − 2); 16/7 > 2, so both logs exist

Lecture 7, Question 2: how long for the current to fade?

The current falls from I to i following $i = Ie^{-t/CR}$. Find an expression for the time t. The unknown is in the power of e, so we'll need **ln**, the undo button for e. Unwrap one layer per line:

$$i = Ie^{-t/CR}$$

$$i = \frac{I}{e^{t/CR}}$$ ← negative power = 1 over (Chapter 9)

$$e^{t/CR} = \frac{I}{i}$$ ← swap i and the e-part (× e^(t/CR), ÷ i)

$$\frac{t}{CR} = \ln \frac{I}{i}$$ ← ln undoes e: ln(e^□) = □

$$t = CR \ln \frac{I}{i}$$ ← × CR

Put numbers in (Chapter 21's question): with $CR = 2$ seconds, how long until the current is 10% of the start? Then $\frac{I}{i} = 10$:

$$t = 2 \ln 10 \approx 2 \times 2.303 = 4.61 \text{ s}$$

Check: $e^{-4.61/2} = e^{-2.303} \approx 0.10$ ✓.

Lecture 7, Question 3: resisted motion

$v = u(1 - e^{-kt})$ gives the velocity v at time t from the starting velocity u. Find t. (The slides do it two ways; this is the shorter, version 3b.)

$$v = u - ue^{-kt}$$ ← expand the bracket

$$v = u - \frac{u}{e^{kt}}$$ ← negative power = 1 over

$$ve^{kt} = ue^{kt} - u$$ ← × e^(kt) every term

$$u = (u - v)e^{kt}$$ ← gather the e-terms: factorise (Chapter 12)

$$e^{kt} = \frac{u}{u - v}$$ ← ÷ (u − v)

$$kt = \ln \frac{u}{u - v}$$ ← ln undoes e

$$t = \frac{1}{k} \ln \frac{u}{u - v}$$ ← ÷ k

Test: $u = 10$, $k = 0.5$, $v = 5$: $t = 2 \ln 2 \approx 1.386$. Back in the original: $10(1 - e^{-0.693}) = 10(1 - 0.5) = 5$ ✓.

Mistakes to avoid

✗ Common mistakes

✗ $\log(x+3) = \log x + \log 3$

the law is for a product, not a sum

✓ $\log(3x) = \log x + \log 3$

✗ $\dfrac{\log a}{\log b} = \log \dfrac{a}{b}$

a quotient of logs isn't a log

✓ $\log \dfrac{a}{b} = \log a - \log b$

✗ $\log_2 x + \log_3 x = \log_6 x$

the laws need the same base

✓ change base first

✗ $(\log x)^2 = 2 \log x$

the power must be inside the log

✓ $\log x^2 = 2 \log x$

✗ $\ln x = 2 \Rightarrow x = \ln 2$

undo ln with e, not with ln

✓ $x = e^2$

✗ $x = -5$ in $\log x$

check answers in the original

✓ reject: no log of a negative

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Definition	$y = \log_b x \iff b^y = x$ $b > 0$, $b \neq 1$; only $x > 0$ has a log
Special values	$\log_b b = 1, \qquad \log_b 1 = 0$
Undo	$\log_b(b^x) = x, \qquad b^{\log_b x} = x$
ln and e	$\ln(e^x) = x, \quad e^{\ln x} = x, \quad \ln e = 1, \quad \ln 1 = 0$ $\ln = \log_e$, $\log = \log_{10}$
Product law	$\log(ab) = \log a + \log b$ same base throughout
Quotient law	$\log \dfrac{a}{b} = \log a - \log b$
Power law	$\log a^n = n \log a$
Reciprocal	$\log \dfrac{1}{a} = -\log a$
Change of base	$\log_a N = \dfrac{\log_b N}{\log_b a}$
Unknown in a power	power alone, log both sides, power law, solve
Log equation	one log each side or log = number; rewrite as a power; reject roots that log a number ≤ 0
No laws for	$\log(a+b)$, $\log a \times \log b$, $\frac{\log a}{\log b}$

Counting multiplications: your turn

Answers in grey.

What is it?

1. $\log_3 9$
2. $\log_2 32$
3. $\log_{10} 0.001$
4. $\log_6 1$
5. Write $5^3 = 125$ as a log
6. Write $\log_2 64 = 6$ as a power

1. 2 2. 5 3. −3 4. 0 5. $\log_5 125 = 3$ 6. $2^6 = 64$

Laws

7. Write as one log: log 4 + log 5
8. Write as one log: 3 log 2 − log 4
9. Expand: $\log(x^3y/z)$

7. log 20 8. log 2 (since 8/4 = 2) 9. 3 log x + log y − log z

Solve

10. $3^x = 81$
11. $5^x = 40$ (2 d.p.)
12. $e^x = 20$ (2 d.p.)
13. How many years for money at 8% to double?
14. Make t the subject: $N = N_0e^{(-\lambda t)}$

10. 4 11. 2.29 12. ln 20 ≈ 3.00 13. log 2 / log 1.08 ≈ 9.0 years 14. $t = (1/\lambda)\ln(N_0/N)$

Find the unknown

15. $\log_2 x = 6$
16. ln x = 3 (2 d.p.)
17. log x + log(x − 3) = 1
18. 5^(2x + 1) = 30 (3 d.p.)
19. $2^x = 5^{(x - 1)}$ (3 d.p.)
20. Find k if 40 = 100e^(−2k) (3 d.p.)

15. $x = 64$ ($x = 2^6$) 16. $x = e^3 \approx 20.09$ 17. $x = 5$ ($x^2 - 3x - 10 = 0$; reject $x = -2$) 18. $x = 0.557$ ($2x + 1 = \log 30 / \log 5 = 2.113$) 19. $x = 1.756$ ($x = \log 5 / (\log 5 - \log 2)$) 20. $k = 0.458$ ($e^{(-2k)} = 0.4$, $k = -\ln 0.4 / 2$)